Quickstart Guide to the DCC Decoder Test Board

This quick start guide is for the DCC Decoder Test Board (aka Sender Board), Series 3, Revision E. The board has the exact same circuit as the previous Series 2 board, but uses surface mount components instead of through-hole components. The ISA card-edge connector has also been removed.

Additional parts/components required:

- Technologic Systems TS-5300 PC-/104 Computer with Embedded DOS
- 5-volt DC supply for TS-5300
- Compact Flash card for Drive C:
- Serial ribbon cable for COM2
- Null Modem/USB Serial Converter
- Ribbon Cable for DB25F

The Series 3 Sender Board was designed to be part of an entire DCC Decoder Test Platform. The plan is to include the computer, timing board (i.e. Sender board), and power station in one package.

The PC/104 computer includes DOS in ROM, and typically boots from the internal Drive A. The DOS console is redirected to COM2, since the computer does not include a video board or a keyboard connection.

Jumpers on the TS-5300 select the serial baud rate for the console, 9600 baud without the jumper, and 115200 baud with the jumper installed. Refer to the TS-5300 documentation for more details.

1. Connect the Serial Ribbon cable to COM2, the null modem, gender changer, and USB to Serial converter to any host computer (PC, Mac, etc.)

2. Plug a compact flash card into the socket.

3. Install the Sender board as the top board in the PC/104 Stack.

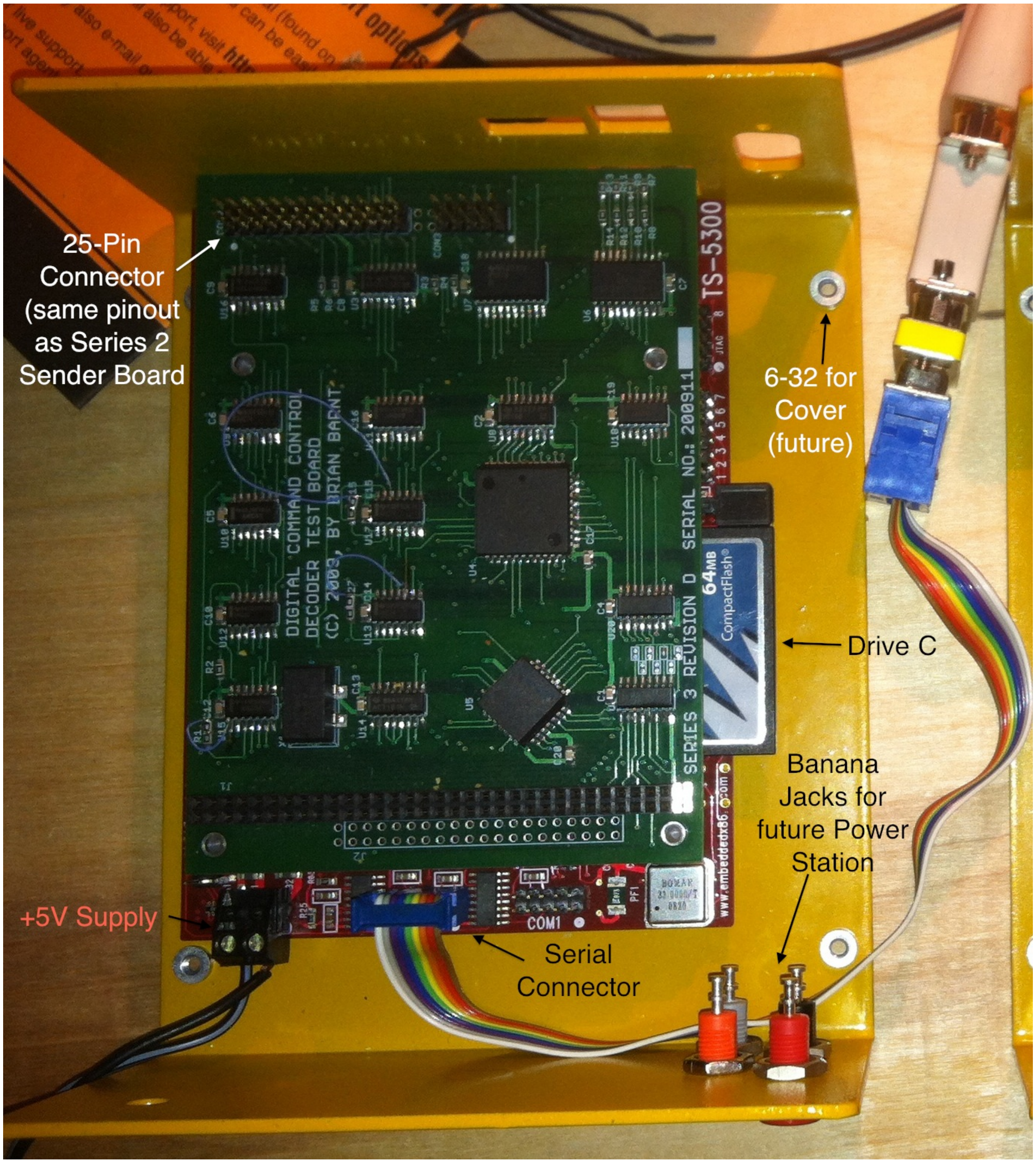

Figure 1: Series 3 Rev D Sender board installed on a TS-5300 with serial cable/null modem/gender changer/USB-Serial interface.

4. Add the 25-pin ribbon cable. Plug the interface (same as Series 1 and Series 2) into the DB25F connector.

The Series 3 also includes the same DB9F 9-pin auxiliary connector; this connector is typically not used. If testing, note that the orientation is 180° from

5. Connect +5V to the removable power connector on the TS-5300.

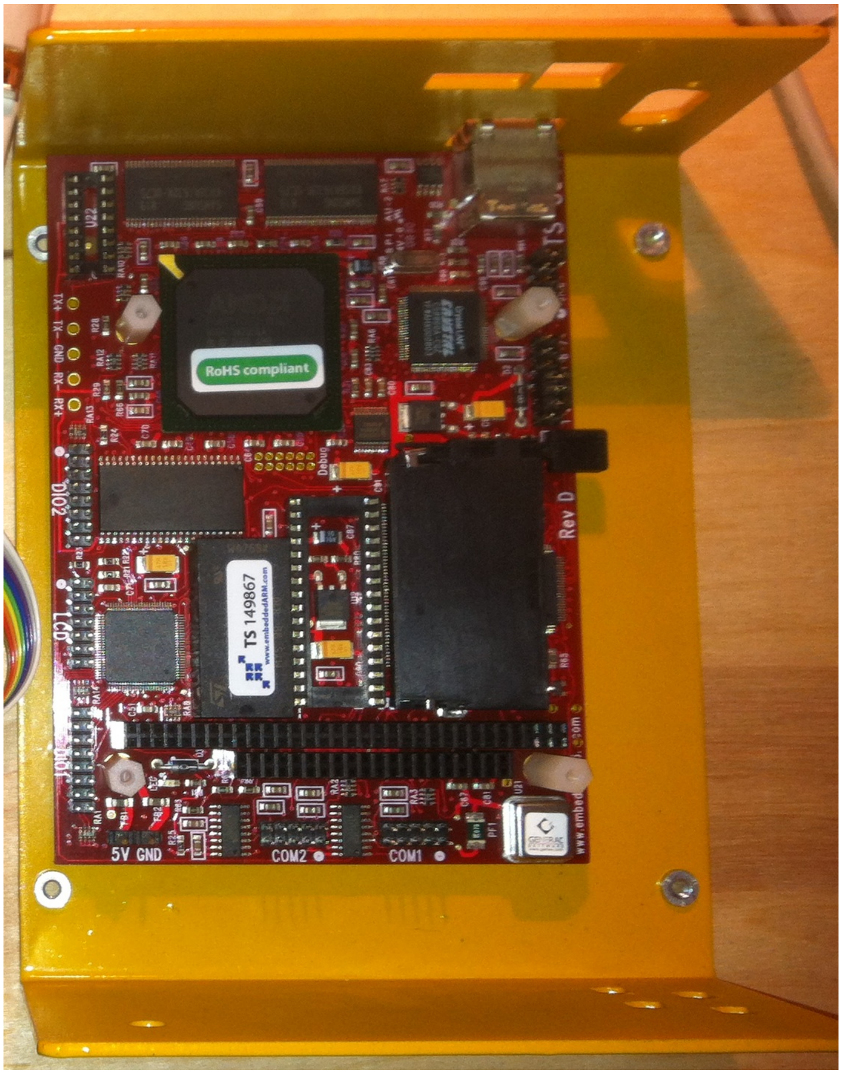

Figure 2: TS-5300 without Sender 3 board installed.

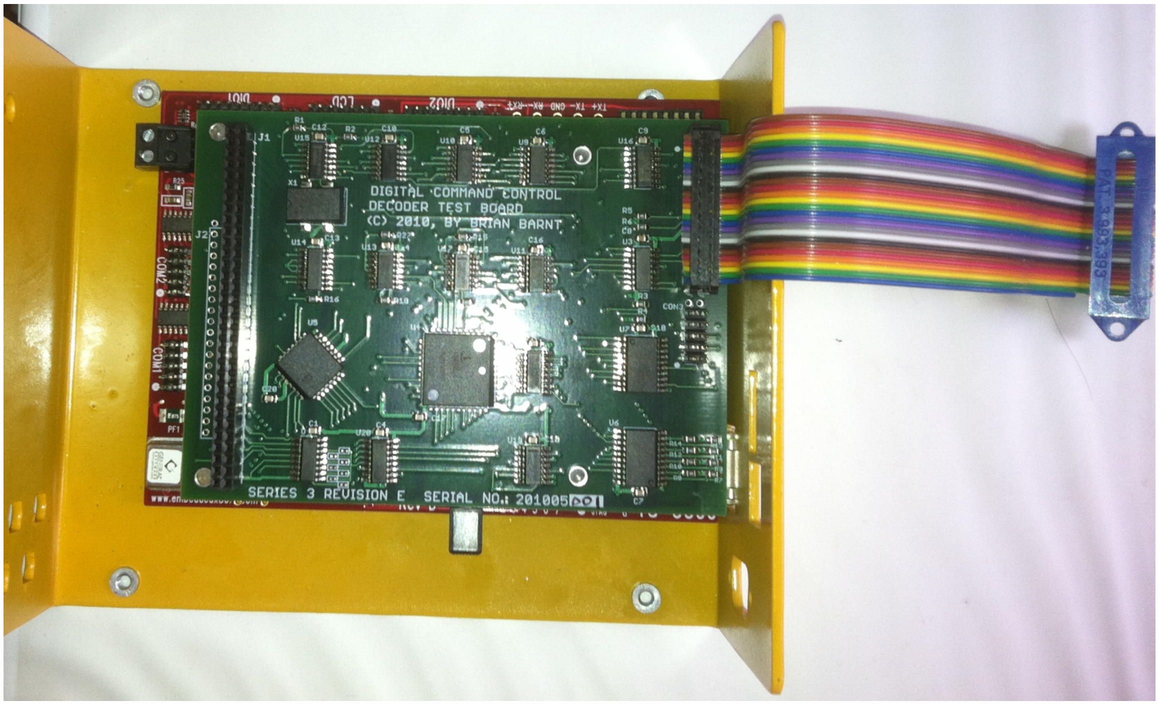

Figure 3: Sender 3 installed on TS-5300 with DB25F Ribbon Cable.